

# Tyler Zeng

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## Education

St. Robert Catholic High School — IB Diploma Programme

Grade 12 | 97% Average

## Experiences

### Sunnybrook Research Institute | Engineering Research Intern

July 2026 - Present

- Developed and calibrated a custom robotic arm system to automate hydrophone data collection through skull phantoms for ultrasound propagation modeling and targeting improvement.
- Implemented **ROS** and **inverse kinematics** for repeatable hydrophone positioning across skull regions and angles.
- Designed an experimental control interface to streamline robotic manipulation, pose selection, and data collection.

### Science Fair Club | Co-Founder & Co-President

Oct 2025 – Present

- Co-founded and scaled a **50+** member student club, overseeing an executive team across various operations.
- Designed and delivered workshops on research design, experimentation, and scientific presentation.

### VEX Robotics Team 3004A | Design & Build Lead

Mar 2024 – Present

- Led the iterative design and construction of competition robots, coordinating team timelines and strategies.
- Mentored new members in mechanical design and **CAD**, ensuring technical consistency and quality across robots.
- Adapted match strategies in real time during tournament play, contributing to multiple championship placements.

## Projects

### NEXUS: A Novel Deafblind Communication Device [5x National Awards]

Python, C, Embedded Systems 🔗

- Designed a wearable accessibility device that converts conversations into speaker-attributed transcripts, which can be accessed through accessible formats for deafblind individuals.
- Constructed using **Raspberry Pi 5** and **STM32**, with a custom 6-microphone array and a modular interface that supports several methods of accessible output (i.e. braille, enlarged text, and haptics).
- Integrated a novel, dynamic speaker identification framework that surpasses state-of-the-art methods for real-time, far-field environments, fusing spatial localization (**TDOA**) and vocal feature extraction (**Pyannote**).

### Fluctus: An Accessible Digital Hearing Aid [National Finalist]

Python, SciPy, NumPy 🔗

- Prototyped a fully digital hearing aid application with real-time adaptive amplification, featuring a **10-band equalizer**, noise suppression (**DeepFilterNet**), vocal restoration (**VoiceFixer**), and tinnitus support.
- Iterated multiple software configurations to remove reliance on hardware, lowering barriers to hearing treatment.

### Longitudinal Multi-Omics Modeling for Early Risk Stratification in IBD

Python, PyTorch, Scikit-learn 🔗

- Designed a dual-encoder **PyTorch** deep learning model predicting inflammatory bowel disease (IBD) progression using longitudinal multi-omics data from **130 patients** and **1,600+ samples**.
- Built a **Gaussian** mixture model identifying three molecular Crohn's disease subgroups for treatment stratification.
- Identified a high-risk subgroup (**10%** of patients) with elevated inflammatory markers, revealing actionable insights.

### SIGMA: Sign Identification with Guided Machine Assistance

Python, OpenCV, YOLOv8, Figma 🔗

- Built a multilingual AI road sign recognition system, leveraging smartphone cameras and **YOLOv8** and a **custom dataset**, achieving **95%** real-time detection accuracy.
- Planned expansion into a fully integrated mobile application for real-time deployment.

## Awards

- Science Fair** — Nationals: Youth Can Innovate Grand Award, Silver Medal, Best in Category, Canadian Acoustical Association Award, \$10,000+ in scholarships, 2x Finalist | Regionals: 2x Gold Medal, 2x Best in Category
- VEX Robotics** — Worlds Qualifier 2026, Provincial Champion, 4x Performance Awards, 6x Judged Awards

## Skills

**Technologies:** Python, C, Embedded Systems, Fusion 360, OnShape, Git